Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Pure 1 – Further Coordinate Geometry

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find the equation of the pair of lines joining the origin to the intersection of $x^2 + y^2 = 1$ and $x + y = 1$. **[4]**

2. Find the angle between the lines $y = 2x + 1$ and $y = 3x - 2$. **[3]**

3. Find the equation of the bisectors of the angles between $y = x$ and $y = 2x$. **[4]**

4. The point $P(t)$ on the parabola $y^2 = 4ax$ has coordinates $(at^2, 2at)$. Find the chord joining t_1 and t_2. **[3]**

5. Find the condition that the chord $t_1 t_2$ of the parabola passes through the focus. **[3]**

6. Find the locus of the midpoint of chords of the parabola $y^2 = 4ax$ which pass through $(a, 0)$. **[5]**

7. Find the equation of the polar of the point (x_1, y_1) with respect to the circle $x^2 + y^2 = a^2$. **[3]**

8. Show that the polar of a point on the director circle is a diameter. **[4]**

9. Find the combined equation of the pair of tangents from (x_1, y_1) to $x^2 + y^2 = a^2$. **[4]**

10. Parametric equations $x = ct$, $y = \frac{c}{t}$ represent a rectangular hyperbola. Find the equation of the tangent at t. **[4]**

11. Find the asymptotes of the hyperbola $xy = c^2$. **[2]**

12. Find the equation of the normal to $xy = c^2$ at t. **[3]**

13. The circle $x^2 + y^2 + 2gx + 2fy + c = 0$ has centre and radius. State them. **[2]**

14. Find the condition that the line $y = mx + c$ is tangent to $x^2 + y^2 = a^2$. **[3]**

15. Find the length of the tangent from (x_1, y_1) to $x^2 + y^2 = a^2$. **[3]**

Section B – Hard (Q16–25)

16. Find the locus of the intersection of perpendicular tangents to the parabola $y^2 = 4ax$. **[5]**

17. Show that the feet of the perpendiculars from the focus to the tangents of a parabola lie on the tangent at the vertex. **[5]**

18. Find the equation of the chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose midpoint is (x_1, y_1). **[4]**

19. Find the condition for the line $lx + my = 1$ to be tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. **[4]**

20. For the rectangular hyperbola $xy = c^2$, show that the chord joining t_1 and t_2 has equation $x + t_1 t_2 y = c(t_1 + t_2)$. **[4]**

21. Find the area of the triangle formed by the tangents at t_1, t_2, t_3 on $xy = c^2$. **[5]**

22. Find the polar of the point (x_1, y_1) w.r.t. the parabola $y^2 = 4ax$. **[3]**

23. Show that the circle on a focal chord of a parabola as diameter touches the tangent at the vertex. **[5]**

24. Find the coordinates of the centre of the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$. **[4]**

25. Reduce $x^2 - xy + y^2 - 2x - 2y = 0$ to standard form by translation/rotation if needed. **[6]**

End of Worksheet • *Show all working clearly.*